

types of how why a claim follows from accepted starting points.

FOUNDATIONS

A

PROPOSITION SYMBOLS AND STATEMENTS

LOGIC SYMBOLS AND STATEMENTS

In everyday examples, treat each "if ... then ..." sentence as an assumed premise.

RULES OF INFERENCE



Modus Ponens or Rules of Inference

p

Simplification

RULES OF INFERENCE

p

Modus Tollens

RULES OF INFERENCE

p

Hypothetical syllogism RULES OF INFERENCE

p

Disjunctive Syllogism

RULES OF INFERENCE

p

◦ all shapes are squares ◦ rectangles; $n = 4$ has 4 sides

BUILDING A PROOF WITH INFERENCE RULES



◦ circle
◦ ellipse
◦ hexagon
◦ pentagon
◦ triangle

Compare each proposed conclusion with what the premises actually allow.

FALLACIES

V

UNDERSTANDING AN IMPLICATION

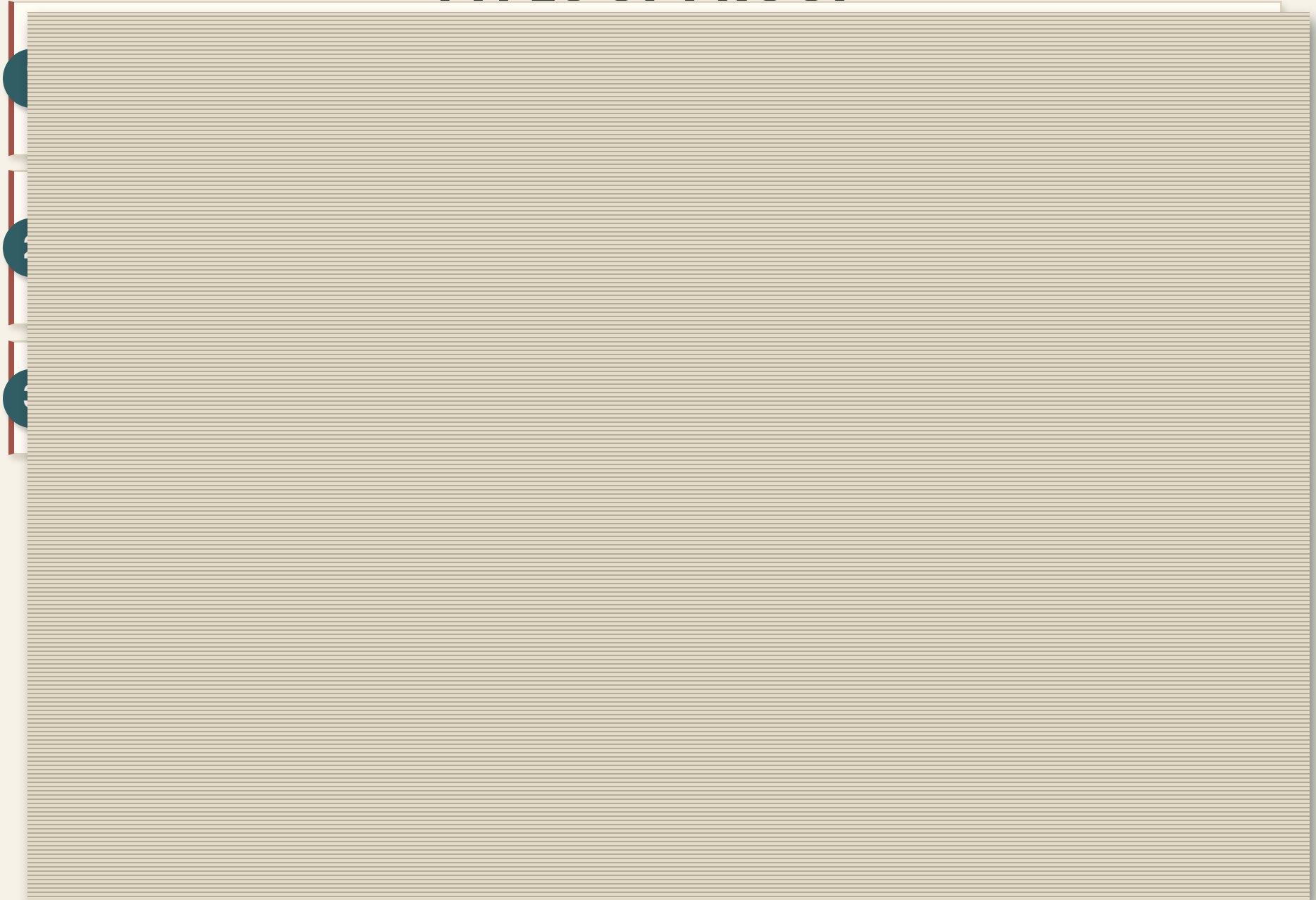
How do you verify the d. Here
is the hypothesis and is the

Try all four cases

c
T
is
W
n
q
A
p
c

Three introductory approaches over the integers:

TYPES OF PROOF



FOR EVERY INTEGER x , IF 6 DIVIDES x , THEN 3 DIVIDES x .

G
a

GUIDED PROOF PRACTICE

For every integer n , if 4 divides n , then 2 divides n .

Hint: Consider the definition of divisibility. Can you find a counterexample?

So

Try

CHALLENGES

Given p and q are given. Which rule gives q ?

A large, empty, light beige rectangular area, likely a workspace for writing or drawing. It is positioned below the header and occupies most of the page.

State the claim and identify its assumptions.

THE FINAL LINE

THANK YOU

...

Q

A